

Section 7 Summary - First Machine Learning Model

Core workflow

Training data teaches the model patterns.

Testing data checks how well the model performs on unseen examples.

Features are numeric representations of raw input data.

For text, CountVectorizer converts words into numeric columns.

The trained model can then predict labels for new input.

Important terms

Accuracy: the percentage of correct predictions.

Prediction: the model output for a new example.

Vectorizer: the preprocessing step that converts text into numbers.

Model: the trained algorithm used for predictions.

C# mapping

Think of the vectorizer and model as a reusable pipeline.

The fitted vectorizer must be reused for future inputs, just like a trained model must be reused consistently.